BSDS (2024 Semester V): Regression Techniques

Quiz 2

NAME :	ROLL :

Time: 50 minutes **Total marks:** 20

Answer all questions. Give explanation to each step used to solve the problems.

1. Let $\underset{\sim}{\theta}$ be a q-dimensional parameter and $\widehat{\underset{\sim}{\theta}}$ be an estimator of $\underset{\sim}{\theta}$. Define the MSE of $\widehat{\underset{\sim}{\theta}}$ as $\mathcal{MSE}(\widehat{\underset{\sim}{\theta}}) = \mathbb{E}\|\widehat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}\|^2$. Express the $\mathcal{MSE}(\widehat{\underset{\sim}{\theta}})$ in terms of bias and variance of $\widehat{\underset{\sim}{\theta}}$. Give proper justification. [3]

2. Let Y be a response variable, and $\{x_1, \ldots, x_p, z_1, \ldots, z_q\}$ be potential predictors. Based on n observations, let $\underset{\sim}{Y}$ be the response vector and $\{\underset{\sim}{x}_1, \ldots, \underset{\sim}{x}_p, \underset{\sim}{z}_1, \ldots, \underset{\sim}{z}_q\}$ be the predictor vectors.

   Consider two sets of models, and two situations A. and B. as follows:

$$\underset{\sim}{Y} = \beta_0 \underset{\sim}{1} + \beta_1 \underset{\sim}{x}_1 + \ldots + \beta_p \underset{\sim}{x}_p + \gamma_1 \underset{\sim}{z}_1 + \ldots + \gamma_q \underset{\sim}{z}_q + \underset{\sim}{\epsilon}^\star; \quad \underset{\sim}{\epsilon}^\star \sim N(\underset{\sim}{0}, \sigma^2 I). \tag{1}$$

$$\underset{\sim}{Y} = \beta_0 \underset{\sim}{1} + \beta_1 \underset{\sim}{x}_1 + \ldots + \beta_p \underset{\sim}{x}_p + \underset{\sim}{\epsilon}; \quad \underset{\sim}{\epsilon} \sim N(\underset{\sim}{0}, \sigma^2 I). \tag{2}$$

   A. Model (1) is true but you work with the model (2).

   B. Model (2) is true but you work with the model (1).

   Let the OLS estimator of $\underset{\sim}{\mu} = E(\underset{\sim}{Y})$ be $\underset{\sim}{\widehat{\mu}}_{\text{OLS}}$. Find the MSEs of $\underset{\sim}{\widehat{\mu}}_{\text{OLS}}$ in situations A. and B. [4 + 4]

3. Consider the normal linear regression model (2). Let VIF_j be the variance inflation factor of the j-th predictor, $j = 1, \ldots, p$, and $\kappa(X)$ be the condition number. Show that

$$\max_{j=1,\ldots,p} \text{VIF}_j \leq \kappa^2(X).$$

[5]

4. Consider the normal linear regression model (2). Let $\widehat{Y}_{i(-i)}$ be the fitted value of Y_i when the regression equation is estimated without using the i-th observation. Find the distribution of the leave-one-out cross validation error, i.e., $Y_i - \widehat{Y}_{i(-i)}$. [4]

Some Useful Results. [1] Let A be a block matrix of the form $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$, and A_{11} is invertible. Then

$$A^{-1} = \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}S^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}S^{-1} \\ -S^{-1}A_{21}A_{11}^{-1} & S^{-1} \end{pmatrix},$$

where $S = A_{22} - A_{21}A_{11}^{-1}A_{12}$.

[2] Let A be a $q \times q$ symmetric matrix, then for $\underset{\sim}{a} \in \mathbb{R}^q$,

$$\lambda_{\min}(A) \;\leq\; \inf_{\underset{\sim}{a} \neq \underset{\sim}{0}} \frac{\underset{\sim}{a}^\top A \underset{\sim}{a}}{\underset{\sim}{a}^\top \underset{\sim}{a}} \;\leq\; \sup_{\underset{\sim}{a} \neq \underset{\sim}{0}} \frac{\underset{\sim}{a}^\top A \underset{\sim}{a}}{\underset{\sim}{a}^\top \underset{\sim}{a}} \;\leq\; \lambda_{\max}(A),$$

where $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$ are the maximum and minimum eigenvalues of A.

Rough work